

de Blame 66 lines (50 loc) · 1.72 KB

Raw

```
1 Q1. Deadlock Debugging
2
3 Cause of Deadlock
4 •P1 locks R1 and waits for R2.
5 •P2 locks R2 and waits for R1.
6 •Each process waits for the resource held by the other (circular wait), causing a deadlock.
7
8 Optimization:
9 Acquire resources in the same order in both processes.
10
11 Corrected Pseudo Code
12 Process P1:
13 lock(R1)
14 lock(R2)
15 critical section
16 unlock(R2)
17 unlock(R1)
18
19 Process P2:
20 lock(R1)
21 lock(R2)
22 critical section
23 unlock(R2)
```

```
24     unlock(R1)
25
26     Q2. CPU Utilization Problem
27
28     Cause:
29     •Processes repeatedly move between Ready and Waiting due to frequent I/O operations.
30     •CPU stays idle while processes wait for I/O, resulting in 30% CPU utilization.
31
32     Optimization:
33     •Use asynchronous I/O or I/O buffering.
34     •Increase multiprogramming so another process can use the CPU while one waits.
35
36     Q3. Memory Bottleneck Debugging
37
38     Cause:
39     •Frequent page faults due to large heap usage (150 MB) and many threads (25 stacks).
40     •Excessive memory accesses reduce performance.
41
42     Optimizations:
43     •Reduce memory usage (optimize heap allocation).
44     •Reduce the number of threads or use a thread pool.
45
46     Q4. Starvation Debugging
47
48     Cause:
49     •Process P5 is in Q3 (lowest priority).
50     •Higher-priority queues (Q1 and Q2) continuously receive processes.
51     •P5 never gets CPU time (starvation).
52
53     Optimization
54     •Apply Aging (gradually increase the priority of waiting processes).
55     •Allocate CPU time periodically to lower-priority queues.
56
57     Q5. Context-Switch Overhead Debugging
58
59     Cause:
60     •Very small time quantum (1 ms).
```

8/4/26, 2:23 PM

Operating-system/CO assessment/CO 2 AT 3 at main · aakifh127-dot-Operating-system

- 61
- Average burst time is 2 ms, so threads are preempted frequently.
- 62
- High I/O activity also causes many context switches.
- 63
- 64
- Optimizations
- 65
- Increase the time quantum (e.g., 4-5 ms).
- 66
- Reduce unnecessary context switches using I/O batching or improve I/O efficiency.